

ROHAN DHANRAJ YADAV

AI/ML Engineer | Generative AI | Multi-Agent Systems | AI Platform Developer

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PROFESSIONAL PROFILE

Results-driven AI/ML Engineer with **7+ years of professional experience** including **4+ years specializing in designing, developing, and deploying production-grade AI systems** across enterprise domains. Deep expertise in **Generative AI, Agentic AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration**, and **Machine Learning**.

Proven track record in leading cross-functional teams of **15+ engineers**, architecting scalable ML/AI pipelines, and delivering high-impact AI applications for **Fortune 500 clients** including Cisco, ITC Fairview, KDP, Omnitrax, PDS, and Marco Technologies. Known for building production systems handling **4,500+ daily API requests**, processing **100K+ daily records**, and achieving **18-38% model performance improvements** in production environments.

Career Metrics

Total Professional Experience	AI/ML Engineering Experience	Agentic AI Experience	Genreative AI Experience	Production AI Projects Delivered	Max Team Size Led	Fortune 500 Clients Served
7+ years	4+ years	1.5 year	3 year	8+	15 engineers	6+

TECHNICAL SKILLS

Programming Languages & Frameworks

Python, Django, Flask, FastAPI, Streamlit

Generative AI & LLM Engineering

Large Language Models (LLMs): OpenAI, Anthropic, Vertex AI, Google Gen AI Platform, Cohere, AI21 Labs

Agentic AI Frameworks: LangChain, LangGraph

Techniques: Prompt Engineering, Structured Outputs Generation, Retrieval-Augmented Generation (RAG), Prompt Chaining, Chain-of-Thought Reasoning, Graph-Based Generation, LLM System Design, LLM Solution Architecture

Agentic AI & Multi-Agent Systems

Agentic AI System Design Patterns: Reflexion, ReAct, Self-Ask, Plan & Execute, Tool Use, Human-in-the-loop, Tree of Thought, Multi-Agent Collaboration, Deep Research

Techniques: Agent Orchestration, LangGraph Workflows, Multi-Agent Systems, Tool Integration, Memory Management, Autonomous Decision-Making, Autonomous AI Agents

Machine Learning & Deep Learning

Scikit-Learn, TensorFlow, Keras, PyTorch, Optuna, HuggingFace Transformers, Model Fine-tuning, Hyperparameter Optimization, PyCaret

Natural Language Processing (NLP)

Spacy, NLTK, Encoder based Transformers, Decoder based Transformers, Encoder-Decoder Transformers, Sentiment Analysis, Named Entity Recognition (NER), Text Classification, Zero-Shot, One-Shot Learning, Aspect Based Sentiment Analysis, Text Generation, Text Summarization, Question Answering, Machine Translation, Paraphrasing

Data Engineering & Databases

Apache Airflow, ETL Processes, SQL DBs (MySQL, PostgreSQL, BigQuery), NoSQL DBs (MongoDB, Cassandra), Vector DBs (Pinecone, Qdrant, LanceDB, Chroma, Pgvector, Milvus, Weaviate)

Analytics & Statistical Computing

NumPy, Pandas, Matplotlib, Seaborn, Plotly, SciPy, Statistical Analysis, Predictive Analytics, Data Visualization

Web Scraping & Data Extraction

BeautifulSoup, Scrapy, Selenium, Playwright, Pydantic Structured Output

DevOps, Cloud & MLOps

Git, GitHub, Docker, CI/CD(Github Workflows, Jenkins), Google Cloud Platform (GCP), Amazon Web Services (AWS), MLOps

WORK EXPERIENCE

1. AI/ML Engineer — Generative AI | Multi-Agent Systems

AI Tech Solutions Ltd. | November 2024 – October 2025 | Full-time

Focus: Architecting production-ready Agentic AI platforms, advanced RAG systems, and multi-agent orchestration

Key Responsibilities & Achievements:

- Led 15-member team building scalable AI systems using Python, FastAPI for backend services and production deployment
- Developed modular ML pipelines (Omodore platform) achieving **38% accuracy improvement** through advanced RAG architecture
- Designed multi-agent orchestration systems using **LangGraph** for complex workflow automation and autonomous decision-making
- Implemented prompt chaining and **chain-of-thought reasoning** mechanisms
- Deployed containerized AI systems using **Docker** with vector databases (Qdrant, LanceDB, Chroma)

- Built automated web scraping systems (Rbetraj) with **Playwright** and **Selenium** for real-time data extraction
- Architected multi-modal virtual agent (GALAMBO) integrating GPT-4 and autonomous tool use

Tech Stack: Python, LangChain, LangGraph, FastAPI, LLMs, RAG, Agent Orchestration, Multi-Agent Systems, Docker, Qdrant, LanceDB, Chroma, Playwright, Selenium

2. Software Engineer — Generative AI | ML | Data Science | Python

Aimlytics Technology | September 2022 – October 2024 | Full-time

Focus: Enterprise-scale Generative AI, NLP, and ML solutions for Fortune 500 clients

Key Responsibilities & Achievements:

- Led team of 8 engineers delivering AI solutions for **Fortune 500 clients**
- Built RAG-based virtual sales assistant (SalesMoji) handling **4,500+ daily API requests**
- Built ML pipelines processing **100K+ daily records** using **Apache Airflow**
- Fine-tuned **BERT** and **RoBERTa** models achieving **18% performance improvement**
- Implemented vector database solutions (Qdrant, Chroma) for semantic search
- Built multi-modal educational AI system (InventtEd)
- Developed multilingual NLP pipeline supporting **15+ languages** (evalmyBRAND)
- Resolved **28 critical production issues** with 100% on-time delivery

Tech Stack: Python, RAG, LangChain, Airflow, BERT, RoBERTa, FastAPI, Flask, TensorFlow, PyTorch, Qdrant, Chroma, GCP

3. Machine Learning Engineer Intern

iNeuron.ai | August 2021 – September 2022 | Internship

Focus: ML pipeline development and optimization

Key Responsibilities & Achievements:

- Built ML pipeline architectures for multi-class classification
- Reduced training time by **35%** using **Optuna**
- Deployed ML applications using Flask and Docker
- Built protein classification system (77 markers, 8 classes)

Tech Stack: Python, Flask, Scikit-Learn, Optuna, MySQL, Cassandra, Docker, AWS

4. Metallurgical Quality Engineer

Bureau Veritas Group | August 2016 – November 2019 | Full-time

Focus: Quality assurance and industrial data analytics

Key Responsibilities & Achievements:

- Conducted metallurgical testing and quality assurance across major steel companies
 - Performed statistical analysis, predictive analysis, and data-driven quality improvement
 - Applied Python (NumPy, Pandas, SciPy, Plotly) for industrial data analysis
 - Managed quality compliance across multiple facilities
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PROJECT SYNOPSIS

1. Project Name: OMODORE

Duration: November 2024 to October 2025

Role: Senior AI Engineer

Description:

Domain: AI Agent Platform

Omodore is an auto-agent platform enabling non-technical users to build domain-specific AI agents without coding expertise. The platform leverages advanced RAG pipelines to deliver accurate, context-aware responses with a 45% accuracy boost.

Key Features:

- No-code AI agent builder for domain-specific applications
- Advanced RAG pipeline with 45% accuracy improvement
- Context-aware intelligent responses using Chain of Thoughts
- Scalable architecture for enterprise deployment
- Automated agent workflows

Roles and Responsibilities:

- Led 15-member team developing Gen AI and Agentic AI solutions
- Architected scalable RAG pipelines using LangChain and LangGraph
- Implemented prompt chaining and Chain of Thoughts methodology for improved reasoning
- Built automated agent building workflows using FastAPI
- Integrated multiple LLMs (Large Language Models) for optimal performance
- Implemented vector database solutions (Qdrant, Milvus, Weaviate) for efficient information retrieval
- Resolved critical production issues ensuring system reliability

Skills Used:

Language: Python

Development Framework: FastAPI

Generative AI: LangChain, LangGraph, GPT, LLMs, Prompt Chaining, Chain of Thoughts, RAG, Agent Building

Cloud & DevOps: Docker

2. Project Name: RBETRAJ

Duration: November 2024 to October 2025

Role: Senior AI Engineer

Description:

Domain: E-commerce Data Intelligence

Rbetraj is an Automated Dropshipping Platform powered by Shop Scan Agent which specializes in e-commerce data extraction with advanced price tracking and competitive analysis capabilities. At the heart of Shop Scan Agent is an Open Source Automated Scraper AI TechCrawler that intelligently scrapes websites, extracts structured data, and provides actionable insights.

Key Features:

- Automated website scraping with intelligent data extraction
- Structured output generation from unstructured web data using Pydantic
- Real-time price tracking across multiple e-commerce platforms
- Competitive analysis and market intelligence
- Multi-source data aggregation

Roles and Responsibilities:

- Developed open-source automated scraping framework [AI Tech Crawler](#)
- Built Shop Scan Agent for e-commerce intelligence
- Implemented GPT-powered and LLM-based structured data extraction
- Created robust scraping pipelines using Playwright
- Designed RESTful APIs using FastAPI for data access
- Integrated LangChain for intelligent data processing workflows
- Implemented Pydantic models for data validation and structured output
- Implemented error handling and retry mechanisms for reliable scraping

Skills Used:

Language: Python

Development Framework: FastAPI

Web Scraping: Selenium, Playwright, BeautifulSoup

Generative AI: GPT, LLMs, LangChain, Pydantic for structured output

Cloud & DevOps: Docker

3. Project Name: GALAMBO

Duration: November 2024 to October 2025

Role: Senior AI Engineer

Description:

Domain: Multi-Modal AI Assistant

GALAMBO is a sophisticated multi-modal virtual agent that combines context-aware GPT responses with advanced image search capabilities, providing users with comprehensive visual and textual information retrieval.

Key Features:

- Multi-modal AI processing (text and images)
- Context-aware conversational responses using LLMs
- Intelligent image search and analysis
- Integration with search engine tools
- Real-time information retrieval

Roles and Responsibilities:

- Architected multi-modal AI system integrating text and image processing
- Implemented context-aware response generation using GPT and LLMs
- Integrated search engine tools for enhanced information retrieval
- Built scalable API infrastructure using FastAPI
- Implemented LangChain workflows for agent orchestration
- Optimized response times and accuracy for production use

Skills Used:

Language: Python

Development Framework: FastAPI

Generative AI: GPT, LLMs, Multi-modal AI, LangChain, Agent Building

Tools: Search Engine Integration

Cloud & DevOps: Docker

4. Project Name: SALESMOJI

Duration: November 2023 to October 2024

Role: Software Engineer (Gen AI | ML | Data Science | PYTHON)

Description:

Domain: Sales and Marketing

SalesMoji harnesses the power of GPT and Large Language Models to understand sales data, answer questions in plain English, and deliver instant, actionable sales intelligence. The platform processes 1M+ daily records through sophisticated Airflow pipelines, transforming raw data into strategic insights.

How it Works: Chat-bot interface where users can type sample questions and receive instant data analytics with visual representations and actionable recommendations.

Key Features:

- AI-Powered Sales Insights using GPT and LLMs
- User-Friendly Conversational Interface
- Customizable Templates and Dashboards
- Real-Time Data Visualization
- Interactive Chat Interaction with 10K+ daily queries
- 89% accuracy in query responses
- 1M+ daily records processing capability

Roles and Responsibilities:

- Led 8-member team delivering custom AI workflows for Fortune 500 clients
- Built RAG-based Virtual Sales Assistant responding to natural language queries
- Achieved 10K+ daily query handling with 89% accuracy using context-aware GPT responses
- Developed RAG pipelines for various sales channels using Vector Databases
- Constructed and orchestrated data pipelines processing 1M+ daily records using Airflow
- Architected Machine Learning pipelines for predictive analytics
- Generated predictive insights for sales forecasting and trend analysis
- Implemented vector database solutions (Qdrant, Milvus, Weaviate) for efficient retrieval
- Applied Prompt Chaining and Chain of Thoughts for complex reasoning
- Built agent workflows using LangChain and LangGraph
- Resolved critical production issues maintaining high system availability

Skills Used:

Language: Python

Development Framework: FastAPI

Databases:

- SQL: MySQL
- NoSQL: MongoDB
- Vector DB: Qdrant, Milvus, Weaviate

Machine Learning: Sklearn, TensorFlow, Keras, PyTorch

NLP: Hugging Face Transformers, Large Language Models (LLMs)

Generative AI: LangChain, LangGraph, GPT, Prompt Chaining, Chain of Thoughts, RAG, Agent Building

Data Orchestration: Airflow

5. Project Name: AISERA

Duration: July 2023 to October 2024

Role: Software Engineer (AI | ML | Data Science | PYTHON)

Description:

Domain: Customer Service and IT Operations Management

Aisera is a leading provider of Generative AI Solutions that helps enterprises boost revenue, improve user productivity, lower operating expenses, and create magical user experiences by automating and personalizing customer support. The platform uses Natural Language Processing (NLP) and Machine Learning (ML) to understand customer queries and provide relevant information or solutions.

Key Features:

- Custom AI models for various enterprise tasks
- AI Service Management (AISM) for automated operations support
- TRAPS Framework for ethical and secure AI practices
- Positive customer feedback on efficiency and automation
- Enterprise-grade workflows for Fortune 500 clients

Roles and Responsibilities:

- Led 8-member team comprising 5 Data Engineers and 3 Software Developers
- Delivered custom AI workflows for Fortune 500 clients including Cisco, ITC Fairview, Omnitrax, PDS, KDP, Marco Technologies
- Fine-tuned BERT models for sentiment analysis achieving 50% performance improvement
- Developed and scheduled system triggers using Airflow for automated operations
- Implemented custom field prediction using advanced BERT architectures
- Architected scalable ML pipelines ensuring reliability and performance
- Implemented vector database solutions (Chroma, Weaviate) for RAG applications
- Integrated LangChain for building conversational AI workflows
- Resolved critical production issues maintaining high system availability
- Handled 28 tickets overall and individually resolved 16 tickets out of 63 tickets assigned to Aimlytics Technologies Workflow team

Skills Used:

Language: Python, Nashorn JAVA

Development Framework: FastAPI

Databases:

- SQL: Postgres
- NoSQL: MongoDB
- Vector DB: Chroma, Weaviate

Machine Learning: Sklearn, TensorFlow, Keras, PyTorch

NLP: Spacy, NLTK, Hugging Face Transformers, BERT, Large Language Models (LLMs)

Generative AI: LangChain, Prompt Chaining, RAG, Agent Building

Data Orchestration: Airflow

Cloud Platform: GCP

6. Project Name: EVALMYBRAND

Duration: September 2022 to October 2024

Role: Software Developer (AI | ML | Data Science | PYTHON)

Description:

Domain: Customer Experience Management

evalmyBRAND is an AI-powered platform designed to turn unstructured customer feedback data into actionable insights, providing a holistic view of customer sentiment and brand perception across multiple channels. The platform supports 15+ languages for comprehensive global brand monitoring.

Key Features:

- Comprehensive Insights and Analytics
- Customer Engagement Tracking
- Aspect-Based Sentiment Analysis with 50% performance gain
- Competitor Analysis and Benchmarking
- Brand Goals Attainment Monitoring
- Intelligent Case Management System
- Multi-language support (15+ languages)

Roles and Responsibilities:

- Developed web application using FastAPI for scalable API services
- Built and orchestrated data pipelines using Airflow for automated data processing
- Managed team of 5 Data Annotators for dataset labeling and quality control
- Created innovative pipeline for mixed code language translation using Google Translate API and LibreOffice
- Fine-tuned RoBERTa model for Aspect-Based Sentiment Analysis achieving 50% performance improvement
- Implemented BERT models for Neural Machine Translation tasks with 27% accuracy improvement
- Optimized model architectures for production deployment
- Integrated multiple data sources for comprehensive brand analysis across 15+ languages
- Implemented sentiment analysis pipelines for real-time brand monitoring

Skills Used:

Language: Python

Development Framework: Poetry, Flask, FastAPI

Databases:

- SQL: Oracle
- NoSQL: MongoDB, Cassandra

Machine Learning: Sklearn, TensorFlow, Keras, PyTorch

NLP: Spacy, NLTK, Hugging Face Transformers, BERT, RoBERTa, Large Language Models (LLMs), Google Translate, LibreOffice Translate, Mixed Code Language Translation

Generative AI: LangChain, Prompt Chaining, RAG

Data Orchestration: Airflow

7. Project Name: INVENTTED

Duration: January 2023 to August 2023

Role: Software Developer (AI | ML | Data Science | PYTHON)

Description:

Domain: Educational Technology

InventtEd™ is a pioneering education platform designed to help educators and students become more efficient, effective, and productive. It provides infinite ways to improve productivity, enhance student learning outcomes, and increase engagement through AI-powered tools.

Key Features:

- Intelligent Text Summarization
- Automated Essay Grading with detailed feedback
- Feedback/Complaint Analysis for institutional improvement
- Unusual Language Detection for content moderation
- Automated Timetable Generator
- Virtual Tutor (Q&A Bot) for student support
- Student Performance Prediction analytics

Roles and Responsibilities:

- Performed advanced prompt engineering to fetch questions and answers from ChatGPT using optimized prompts
- Fine-tuned CLIP (Multi-Modal Vision Transformer) to generate images from text descriptions
- Fine-tuned Flan-T5 model to generate keyword-based MCQs from given educational context
- Designed scalable Phishing Classifier using classical Machine Learning algorithms for email security
- Built multi-modal model for social media content classification (Input: post text + image, Output: category)
- Integrated multiple AI models into cohesive educational platform
- Optimized Transformer models for real-time educational applications

Skills Used:

Language: Python

Development Framework: Poetry, Flask, FastAPI

Databases:

- SQL: Oracle
- NoSQL: MongoDB, Cassandra

Machine Learning: Sklearn, TensorFlow, Keras, PyTorch

NLP: Spacy, NLTK, Hugging Face Transformers, Large Language Models (LLMs), CLIP, Flan-T5

Generative AI: LangChain, Prompt Chaining, RAG, Agent Building, Prompt Engineering

Translation Tools: Google Translate, LibreOffice Translate, Mixed Code Language Translation

Cloud Platform: GCP

8. Project Name: MICE PROTEIN EXPRESSION

Duration: August 2021 to September 2022

Role: Machine Learning Engineer Intern

Description:

Domain: Bioinformatics and Protein Classification

Protein Expression classification models are frequently viewed not only as a challenging task but also as a classification problem that requires balancing accuracy and efficiency in analysis validation due to large-scale datasets.

This project analyzed expression levels of 77 proteins measured in the cerebral cortex of 8 classes of control and Down syndrome mice exposed to context fear conditioning—a task used to assess associative learning. The primary aim was to identify subsets of proteins that are discriminant between the classes, making this a complex multi-class classification problem.

Key Features:

- Multi-class classification across 8 distinct classes
- Analysis of 77 protein expression markers
- Automated ML pipeline for model selection
- Hyperparameter optimization for optimal performance
- 40% training time reduction via Optuna

Roles and Responsibilities:

- Led team of 10 members in developing ML solution
- Projected and architected comprehensive Machine Learning System
- Built automated ML architectures for efficient model development
- Automated Machine Learning pipelines for hyperparameter tuning achieving 40% training time reduction via Optuna
- Implemented model selection algorithms to identify best performing models for production deployment
- Developed data preprocessing pipelines for protein expression data
- Created scalable architecture for handling large biological datasets
- Deployed scalable ML applications with Flask web framework
- Integrated Cassandra DB and MySQL for efficient data management

Skills Used:

Language: Python

Development Framework: Flask

Databases: Cassandra DB, MySQL

Machine Learning: Sklearn (Scikit-Learn), Optuna for hyperparameter tuning, PyCaret **Tools:** Anaconda, Data Pipeline, Artificial Intelligence, Machine Learning

Cloud & DevOps: AWS, Docker

EDUCATION

Degree	Institution	Year
B.Tech. Mechanical Engineering	Biju Patnaik University of Technology, Odisha	2016

GATE Qualified — Credential ID: A2WR3

CERTIFICATIONS

Certification	Issuer	Credential ID
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GATE ME 2016	IIT/NTA	A2WR3
Machine Learning Zoom Camp	DataTalksClub	8F163B
Gold Badge: Python	HackerRank	6920934A49FA
Gold Badge: Problem Solving	HackerRank	113775E95540
Gold Badge: Statistics (10 Days)	HackerRank	—
30 Days of ML Challenge	Kaggle	Rank 69/7,573 teams

ACHIEVEMENTS & AWARDS

- **7+ years** of professional experience including 4+ years in AI/ML engineering
- Achieved **Gold Badges** for Python, Problem Solving, and Statistics on HackerRank
- Secured **Rank 69 out of 7,573 teams** in Kaggle ML Challenge
- Led teams of **8–15 members** across enterprise AI projects
- Delivered **50% model performance improvements** in production systems
- Built systems handling **4,500+ daily API requests**
- Processed **100K+ daily records** via Airflow pipelines
- Achieved **38% accuracy boost** in RAG systems
- Reduced training time by **35%** using Optuna
- Resolved **28 critical production issues**

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